Project Report: UCI Machine Learning Repository Analysis

Tools: Python (pandas, matplotlib, seaborn) · PostgreSQL · HTML/CSS/JavaScript (Chart.js)
Data: 406 datasets from the UCI Machine Learning Repository, 1987 to 2018

1. Executive summary

The UCI Machine Learning Repository is a standard source of practice and benchmark datasets. In this project I cleaned its catalogue data in Python, answered analytical questions about its datasets, attributes and instances with advanced PostgreSQL queries, and built an interactive web dashboard to explore the findings.

The main takeaways:

- **Growth accelerated after 2010.** 244 of the 389 dated datasets (63%) were added from 2010 onward, with the peak in 2016 (40 datasets).
- **Classification dominates.** 228 datasets (56%) are classification tasks. Recommender systems (2) and causal discovery (6) are barely represented.
- **A typical dataset is small, but the range is huge.** The median dataset has about 1,600 rows and 21 attributes, yet the largest has 43.9 million rows.
- **Datasets are getting bigger.** Median size rose from 520 rows in the 1990s to about 2,900 in the 2010s, and the share of Extra Large datasets more than doubled (17% to 37%).
- **Size and complexity are different things.** Rows and attributes are only weakly related, and 30 datasets have more attributes than rows.

2. Objectives

1. Clean and combine the repository's two catalogue tables.
2. Explore the data visually in Python.
3. Answer questions about growth, tasks, size and data types with advanced SQL.
4. Present the results in an interactive dashboard that anyone can open in a browser.

3. Dataset

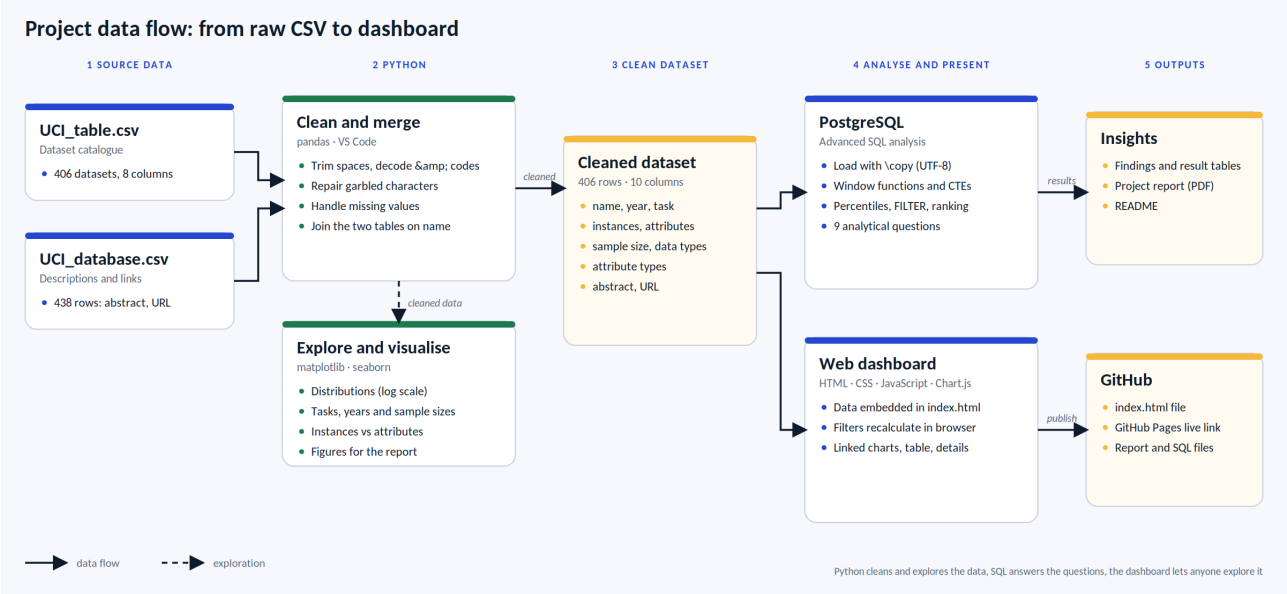
Table	Columns	Rows
`uci_table`	name, data_types, default_task, attribute_types, number_of_instances, number_of_attributes, year, sample_size	406
`uci_database`	name, abstract, identifier_string, datapage_url	438

32 datasets in `uci_database` have no matching row in `uci_table` (no instance counts or years), so they are left out of the analysis. All 406 datasets in `uci_table` matched an abstract and link after cleaning the names.

`sample_size` groups datasets by row count: Small (10 to 100), Medium (101 to 1,000), Large (1,024 to 10,000) and Extra Large (over 10,000).

4. Workflow

1. **Python:** load the raw CSV files, clean them and explore them with charts.
2. **PostgreSQL:** load the cleaned data and run advanced queries to answer the analytical questions.
3. **Web dashboard:** build an interactive page from the cleaned dataset, hosted on GitHub Pages.



5. Data cleaning in Python

Issue found	Action
A stray index column in the table file	Dropped
Inconsistent column names (spaces, capitals)	Renamed to `snake_case`
Extra and double spaces in dataset names, which stopped 11 names matching between the two tables	Trimmed and collapsed spaces
HTML codes such as `&` in names	Decoded
Garbled characters (for example `â€™` for an apostrophe) caused by an encoding mismatch	Repaired and re-saved as UTF-8
Years and counts stored as decimals	Converted to whole numbers, keeping blanks
One dataset with an attribute count of 0	Set to missing
The same attribute type combination written in two orders	Standardised
Two datasets whose names differ only by capital letters (Air Quality and Air quality)	Kept as separate datasets
Missing values: 17 years, 30 attribute counts, 14 data types, 46 attribute types	Left blank rather than imputed, and reported on the charts

6. Exploratory analysis in Python

I used matplotlib and seaborn in VS Code to look at the data before moving to SQL. The charts covered:

- the distribution of instances and attributes (on log scales, because a few huge datasets dominate)
- the number of datasets per task
- the number of datasets added per year
- instances against attributes
- a comparison of size across tasks

What the exploration showed: the data is highly skewed, so averages are misleading. The average dataset has about 400,000 rows and 26,000 attributes, while the median has only about 1,600 rows and 21 attributes. I therefore used medians in the SQL analysis and the dashboard.

7. SQL analysis

All queries use standard PostgreSQL syntax and run against `uci_table`.

Q1. Growth per year: cumulative total, year-on-year change and 3-year rolling average

```
WITH y AS (
  SELECT year, COUNT(*) AS n
  FROM uci_table
  WHERE year IS NOT NULL
  GROUP BY year
)
SELECT year, n,
       SUM(n) OVER (ORDER BY year) AS cumulative,
       ROUND(100.0 * (n - LAG(n) OVER (ORDER BY year))
             / NULLIF(LAG(n) OVER (ORDER BY year), 0)) AS yoy_pct,
       ROUND(AVG(n) OVER (ORDER BY year ROWS BETWEEN 2 PRECEDING AND CURRENT ROW), 1) AS rolling_3yr
FROM y
ORDER BY year;
```

Findings: Additions were slow and steady through the 1990s (about 3 to 16 a year), almost stopped from 2000 to 2007 (no datasets at all in 2004 and 2005), jumped in 2008 (17, up 240% on 2007), and then rose sharply from 2013 onward. The five busiest years were 2016 (40), 2017 (38), 2014 (37), 2013 (33) and 2015 (26). The rolling average peaked at about 35 datasets a year in 2016 to 2018. Dated datasets total 389.

Q2. Share of datasets by task

```
SELECT default_task,
       COUNT(*) AS datasets,
       ROUND(100.0 * COUNT(*) / SUM(COUNT(*)) OVER (), 1) AS pct
FROM uci_table
GROUP BY default_task
ORDER BY datasets DESC;
```

Task	Datasets	Share
Classification	228	56.2%
Clustering	73	18.0%
Regression	69	17.0%
Other/Unknown	28	6.9%
Causal Discovery	6	1.5%
Recommender Systems	2	0.5%

Findings: More than half the repository is classification. If you are looking for recommender or causal discovery data, there is very little to choose from.

Q3. Median and 90th-percentile size per task

```
SELECT default_task,
       COUNT(*) AS datasets,
       PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY number_of_instances) AS median_instances,
       PERCENTILE_CONT(0.9) WITHIN GROUP (ORDER BY number_of_instances) AS p90_instances,
       PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY number_of_attributes) AS median_attributes
FROM uci_table
GROUP BY default_task
ORDER BY datasets DESC;
```

Task	Datasets	Median instances	90th percentile	Median attributes
Classification	228	1,264	75,390	22
Clustering	73	6,611	432,487	31
Regression	69	1,994	96,169	18.5
Other/Unknown	28	1,012	201,736	10
Causal Discovery	6	5,001,901	14,464,573	128
Recommender Systems	2	44,192	49,376	294

Findings: Classification datasets are typically small, with a median of 1,264 rows. Clustering datasets are about five times larger, and causal discovery datasets are in the millions of rows. Each task has a very long tail: the 90th percentile is far above the median in every group.

Q4. Size mix by task

```
SELECT default_task,
       COUNT(*) AS datasets,
       COUNT(*) FILTER (WHERE sample_size = 'Small')       AS n_small,
       COUNT(*) FILTER (WHERE sample_size = 'Medium')      AS n_medium,
       COUNT(*) FILTER (WHERE sample_size = 'Large')       AS n_large,
       COUNT(*) FILTER (WHERE sample_size = 'Extra Large') AS n_xl,
       ROUND(100.0 * COUNT(*) FILTER (WHERE sample_size = 'Extra Large') / COUNT(*), 1) AS xl_pct
FROM uci_table
GROUP BY default_task
ORDER BY datasets DESC;
```

Task	Datasets	Small	Medium	Large	Extra Large	Extra Large share
Classification	228	15	94	65	54	23.7%
Clustering	73	5	22	14	32	43.8%
Regression	69	5	23	20	21	30.4%
Other/Unknown	28	4	10	2	12	42.9%
Causal Discovery	6	0	1	1	4	66.7%
Recommender Systems	2	0	0	0	2	100%

Findings: Classification and regression datasets are mostly Medium sized. Clustering, causal discovery and the Other/Unknown group lean towards Extra Large datasets, and both recommender datasets are Extra Large.

Q5. Largest dataset for each task

```
SELECT default_task, name, number_of_instances
FROM (
  SELECT default_task, name, number_of_instances,
         ROW_NUMBER() OVER (PARTITION BY default_task
                            ORDER BY number_of_instances DESC) AS rk
  FROM uci_table
) ranked
WHERE rk = 1
ORDER BY number_of_instances DESC;
```

Task	Largest dataset	Instances
Clustering	Heterogeneity Activity Recognition	43,930,257
Causal Discovery	KASANDR	17,764,280
Classification	HIGGS	11,000,000
Regression	Gas sensor array under dynamic gas mixtures	4,178,504
Other/Unknown	QtyT40I10D100K	3,960,456
Recommender Systems	Entree Chicago Recommendation Data	50,672

Q6. Datasets with more attributes than rows

```
SELECT COUNT(*) FILTER (WHERE number_of_attributes > number_of_instances) AS wide_datasets,
       COUNT(number_of_attributes) AS datasets_with_attribute_count
FROM uci_table;

SELECT name, number_of_instances, number_of_attributes, default_task
FROM uci_table
WHERE number_of_attributes > number_of_instances
ORDER BY number_of_attributes DESC
LIMIT 5;
```

Findings: 30 of the 376 datasets with a recorded attribute count (8%) are "wide", with more attributes than rows. The widest is URL Reputation (3.2 million attributes), followed by two gas-sensor datasets and a YouTube video games dataset. These are hard cases for modelling because they invite overfitting.

Q7. Have datasets grown over time?

```
SELECT CASE WHEN year < 2000 THEN '1987-1999'
            WHEN year < 2010 THEN '2000-2009'
            ELSE '2010-2018' END AS era,
       COUNT(*) AS datasets,
       PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY number_of_instances) AS median_instances,
       PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY number_of_attributes) AS median_attributes,
       ROUND(100.0 * COUNT(*) FILTER (WHERE sample_size = 'Extra Large') / COUNT(*), 1) AS xl_pct
FROM uci_table
WHERE year IS NOT NULL
GROUP BY era
ORDER BY era;
```

Era	Datasets	Median instances	Median attributes	Extra Large share
1987 to 1999	106	520	17	17.0%
2000 to 2009	39	2,536	41.5	30.8%
2010 to 2018	244	2,889	24	37.3%

Findings: Median row counts rose more than five-fold from the 1990s to the 2010s. Attribute counts did not follow the same pattern: they peaked in the 2000s and fell back.

Q8. Attribute types

```
SELECT COALESCE(attribute_types, 'Unknown') AS attribute_types,
       COUNT(*) AS datasets,
       ROUND(100.0 * COUNT(*) / SUM(COUNT(*)) OVER (), 1) AS pct
FROM uci_table
GROUP BY 1
ORDER BY datasets DESC;
```

Findings: Real-valued attributes alone are the most common (150 datasets, 36.9%), followed by Integer only (16.3%) and Integer plus Real (14.3%). 86 datasets (21%) include categorical attributes. 11% have no attribute type recorded.

Q9. Data types

```
SELECT TRIM(dtype) AS data_type, COUNT(*) AS datasets
FROM uci_table,
     LATERAL regexp_split_to_table(data_types, ',') AS dtype
GROUP BY 1
ORDER BY datasets DESC;
```

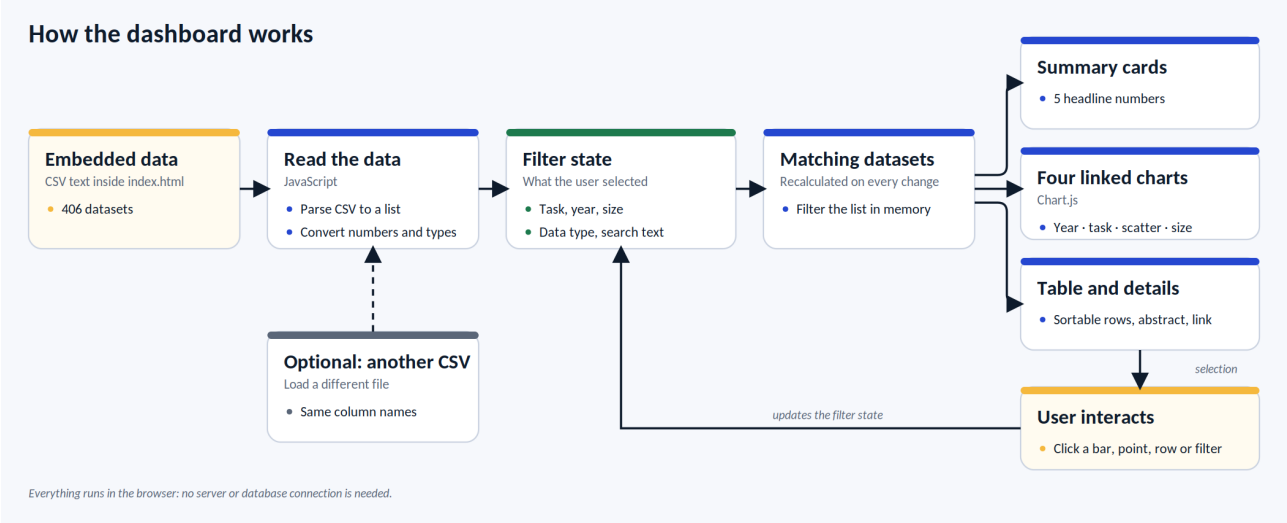
Findings: Because many datasets have several types, the types had to be split into separate rows first. Multivariate appears in 323 of the 392 datasets with a recorded type (82%), followed by Time-Series (76), Text (42), Sequential (41), Univariate (18) and Domain-Theory (16).

Correlation check

The correlation between the logarithms of instances and attributes is about **0.16** (376 datasets), which is weak. More rows does not mean more attributes.

8. Web dashboard

I turned the cleaned dataset into an interactive dashboard, so that people can explore the results without running any code. It is a single `index.html` file that I host on GitHub Pages. I built it with HTML, CSS and JavaScript, using Chart.js for the charts, with AI assistance (Claude).



How it works

- The cleaned and combined dataset is embedded in the page as CSV text. When the page opens, JavaScript reads it into a list of datasets.
- Every filter change recalculates the matching datasets in the browser and redraws all the charts and cards, so no server or database connection is needed.
- A "load another CSV" option lets the page open a new file with the same columns.

Features

Area	What it does
Cards	Total datasets, median instances, median attributes, years covered and largest dataset
Filters	Task buttons, year range, sample size, data type, text search over names and abstracts, and a reset button
Year chart	Datasets added per year with a cumulative line. Clicking a bar filters by that year
Task chart	Datasets by task. Clicking a bar filters by task
Scatter plot	Instances against attributes on log scales. Clicking a point opens its details
Sample size chart	Share of Small, Medium, Large and Extra Large datasets
Table	Sortable. Clicking a row shows the abstract and a link to the dataset

9. Data limitations

1. **A single snapshot.** The data shows what the repository contains, not how often datasets are used or downloaded.
2. **One task per dataset.** `default_task` gives one label, but many datasets suit several tasks.
3. **Missing values.** 17 datasets have no year and 30 have no attribute count, so they are left out of the charts that need those fields.
4. **Skewed sizes.** A few very large datasets dominate averages, which is why I report medians.
5. **Age of the data.** The latest recorded year is 2018, so recent datasets are not included.

10. Conclusions and next steps

The UCI repository has grown mainly since 2010 and is built around small to medium classification datasets, even though its largest datasets run to tens of millions of rows. Datasets have become larger over time, but not more complex in terms of attributes.

Next steps I would take:

1. Add download or citation counts to see which datasets are actually popular.
2. Group datasets by domain (health, finance, text, sensors) using their abstracts.
3. Connect the dashboard directly to the PostgreSQL database instead of embedding the data.

11. Data source

UCI Machine Learning Repository, https://archive.ics.uci.edu/. Dataset descriptions belong to their original authors and donors.